HART Communication Protocol

HART- Highway addressable Remote Transducer

- Introduction
- How it came
- Theory of operation
- Wiring and Installations
- Operation
- Benefits of HART Communication
- Conclusion
- References

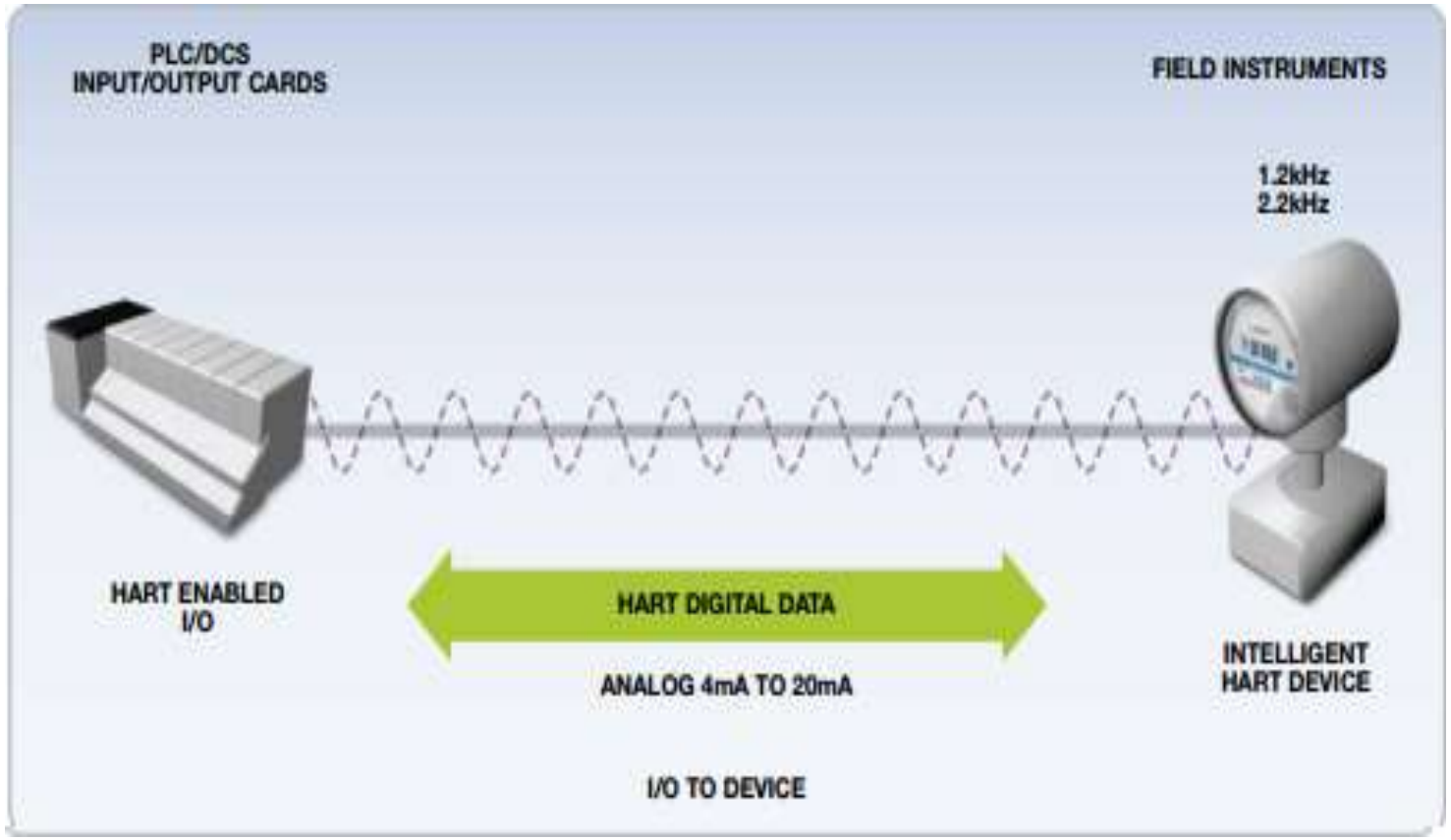
INTRODUCTION

- HART -HIGHWAY ADDRESSABLE REMOTE TRANSDUCER
- Global standard for sending and receiving digital information across analog wires between smart field devices and control or monitoring systems, like PLC and DCS systems.
- Field Devices: pressure, flow, temp. transmitters, valve positioners and actuators.
- Digital networking facility to access any instrument, field device or controller for configuration and diagnostics from any point of the network.
- Digital information utilizes the existing 4 mA to 20 mA network, making it easy to deploy in new as well as existing systems.
- Accurately encode and decode in harsh and noisy conditions.

INTRODUCTION

- It is an open source process control network that supports Hybrid Communication.
- Provides bidirectional digital communication channel over the same 4-20 mA wiring in half duplex mode.
- It uses 1200 baud Frequency Shift Keying (FSK) based on the Bell 202 standard to superimpose digital information on the conventional 4-20 mA analog signal.
- Provide flexibility not available with any other communication technology.
- Analogus to CALLER ID.

HART


PLC/DCS
INPUT/OUTPUT CARDS
FIELD INSTRUMENTS
1.2kHz
2.2kHz
HART ENABLED
I/O
HART DIGITAL DATA
ANALOG 4mA TO 20mA
INTELLIGENT
HART DEVICE
I/O TO DEVICE

How it Came

- Earlier all installed systems in process control industries used 4-20 mA international standard for communicating process variable information between process automation equipments.
- HART Field Communications Protocol extends this 4-20mA standard to enhance communication with smart field instruments, without disturbing the 4-20 mA line.
- The protocol was developed by Rosemount Inc., built off the Bell 202 early communications standard, in the mid-1980s as proprietary digital communication protocol for their smart field instruments. Soon it evolved into HART.
- HART communications Foundation (HCF) was made which takes the responsibility of coordinating and supporting the open protocol standard and manages within this framework the device descriptions of all registered devices.

HART Communication Foundation

- It is a manufacturer-independent, not-for-profit organization encourages widespread use of the HART technology.
- Over five million HART field instruments installed in over 100,000 plants worldwide.



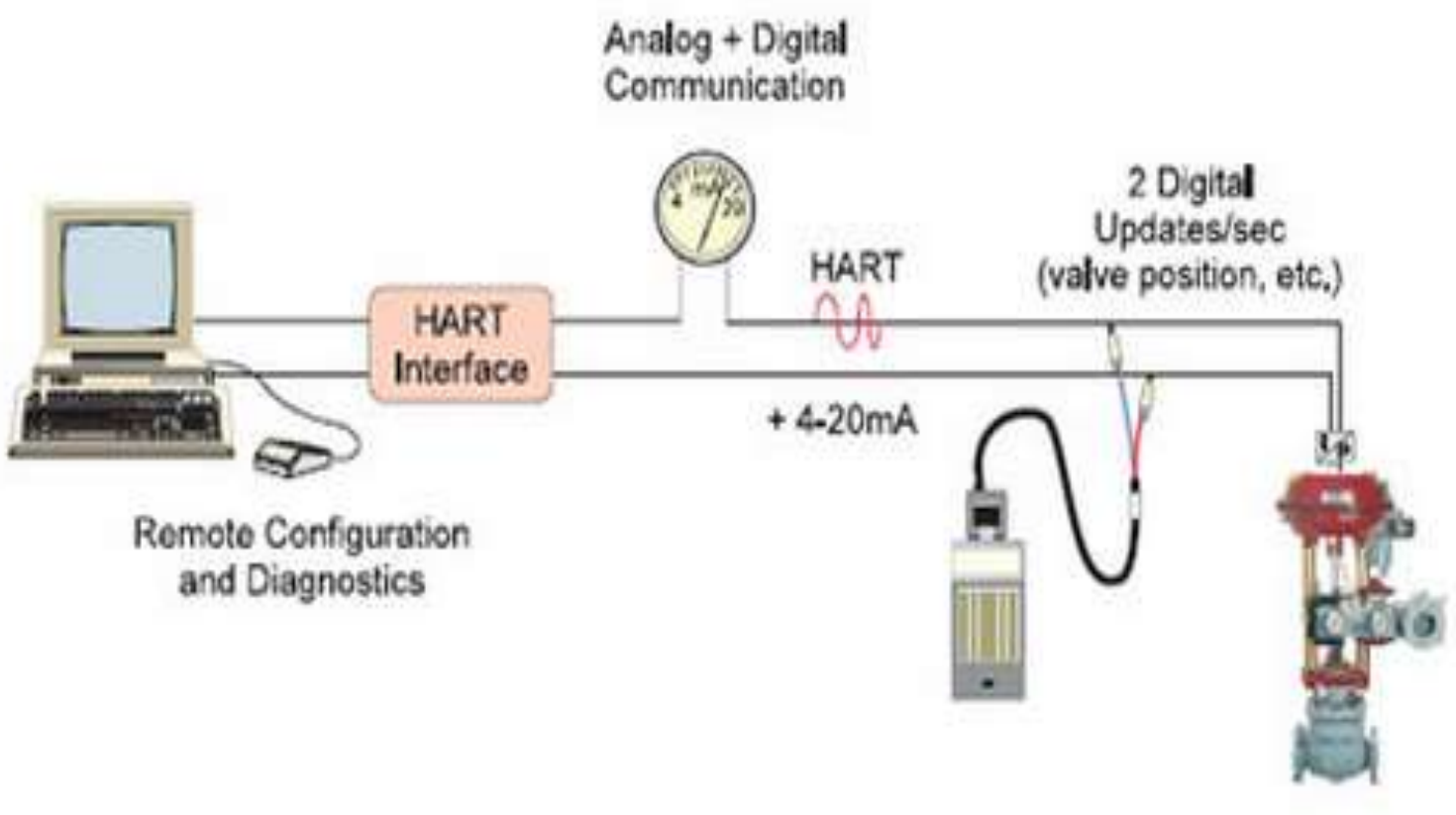
Why HART is globally acceptable?

- Is supported by all of the major vendors of process field instrument
- Preserves present control strategies by allowing traditional 4-20 mA signals to co-exist with digital communication on existing two-wire loops
- Is compatible with traditional analog devices
- Provides important information for installation and maintenance, such as Tag-IDs, measured values, range and span data, product information and diagnostics
- Can support cabling savings through use of multidrop networks
- Reduces operation costs, through improved management and utilization of smart instrument networks.

Theory of Operation

- The HART protocol makes use of the Bell 202 Frequency Shift Keying (FSK) standard to superimpose digital communication signals at a low level on top of the 4-20mA.
- This Signal is sent over conventional wires in process industries.
- The basic principles behind the operation of HART instruments and networks:
  - Communication Modes
  - Frequency Shift Keying
  - HART Networks
  - HART Commands

Demonstration of HART



Theory of Operation

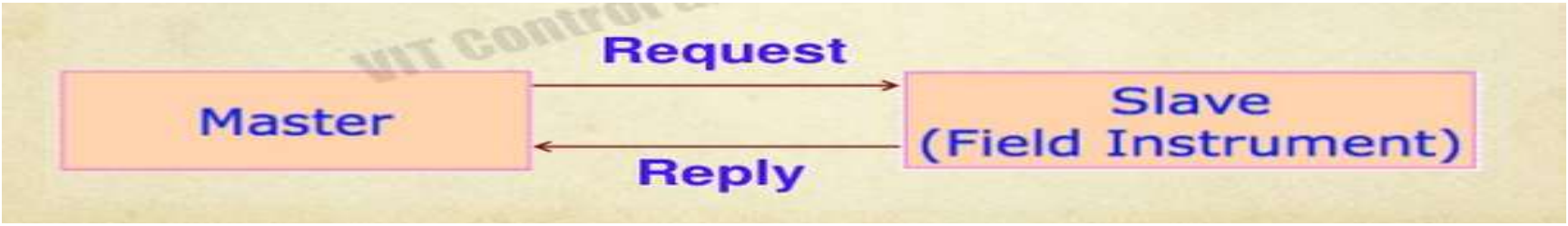
- **COMMUNICATION MODES**

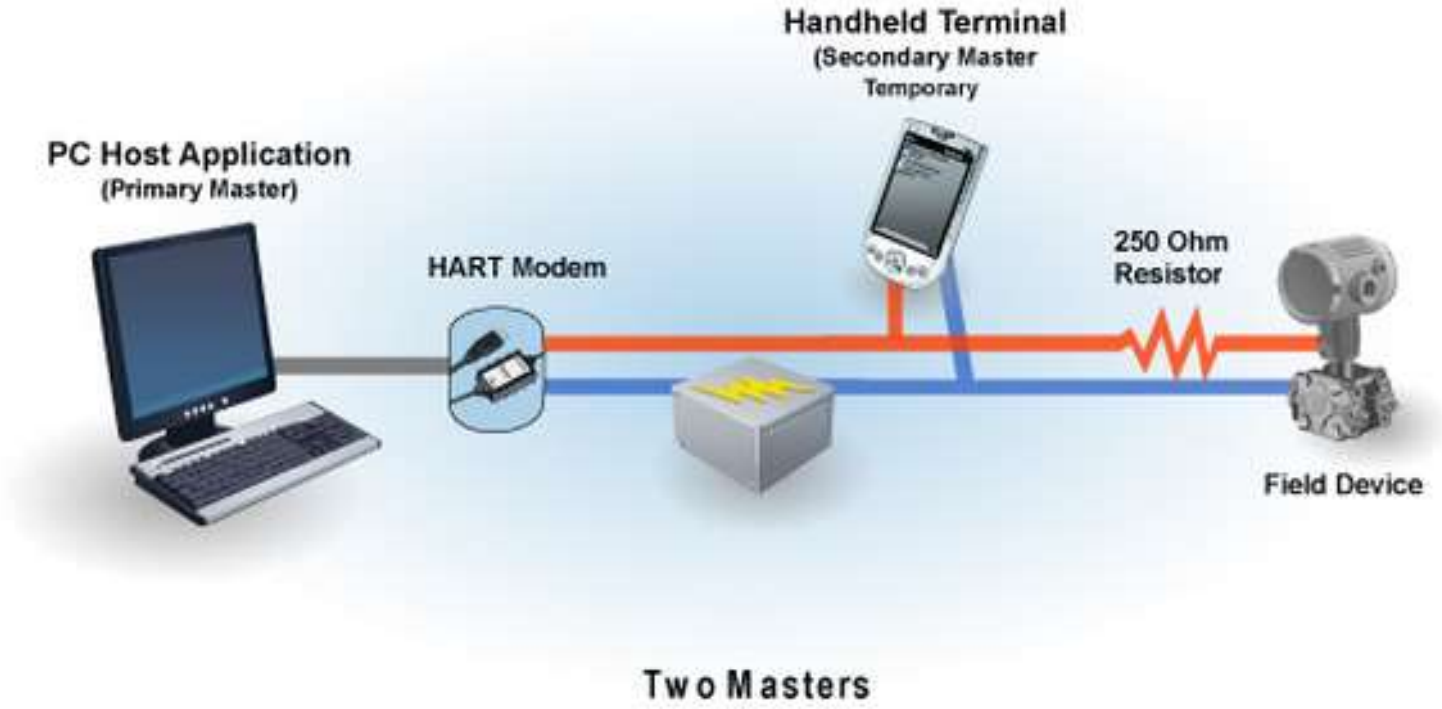
1. **Master Slave Mode**

- It is also called “Poll-Response mode”.
- During normal operation, each slave (field device) communication is initiated by a master communication device (Controller).
- HART provides for up to two masters -primary and secondary.
- The primary master is generally a distributed control system (DCS), programmable logic controller (PLC), or a personal computer (PC).
- The secondary master can be a handheld terminal or another PC. Slave devices include transmitters, actuators, and controllers that respond to commands from the primary or secondary master.

Master Slave mode




Request
Master
Slave
(Field Instrument)
Reply


Handheld Terminal
(Secondary Master
Temporary
PC Host Application
(Primary Master)
HART Modem
250 Ohm
Resistor
Field Device

Two Masters

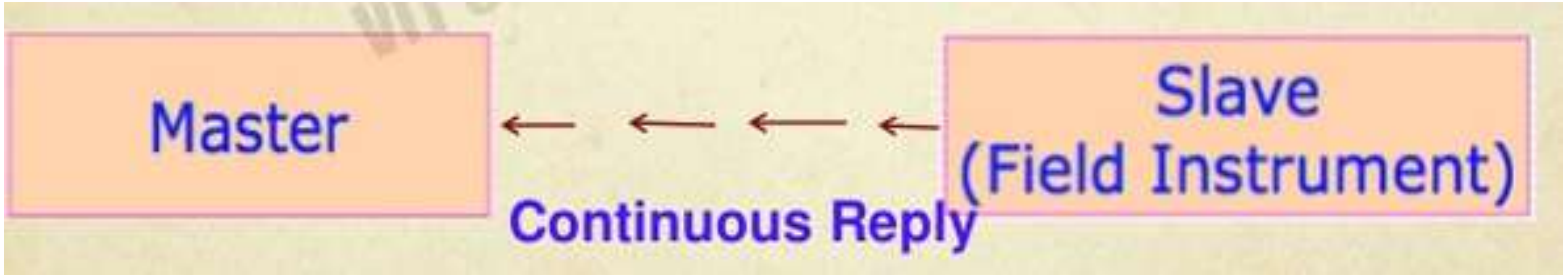
Universal Handheld Communicator



Communication Modes of HART

2. Burst Mode

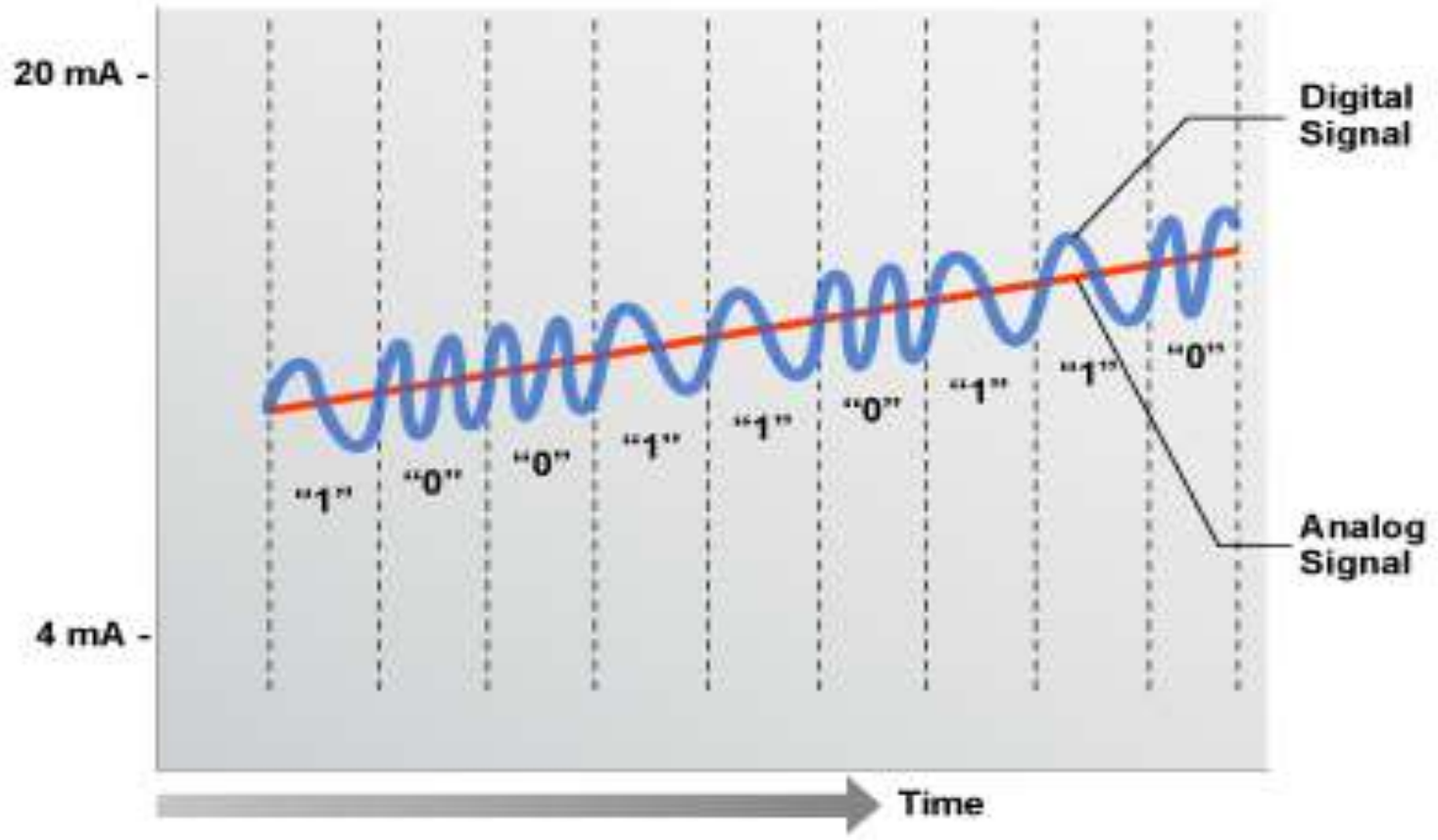
- In burst mode, the master instructs the slave device to continuously broadcast a standard HART reply message (e.g., the value of the process variable). The master receives the message at the higher rate until it instructs the slave to stop bursting. It is also called “Broadcast Mode”.
- It has a maximum communication speed of 3 messages/ second. It is generally used where fast updation of the value of a measured variable is required.



FREQUENCY SHIFT KEYING

- It is based on the Bell 202 telephone communication standard and operates using the frequency shift keying (FSK) principle.
- The digital signal is made up of two frequencies— 1,200 Hz and 2,200 Hz representing bits 1 and 0, respectively.
- Sine waves of these two frequencies are superimposed on the direct current (dc) analog signal cables to provide simultaneous analog and digital communications.
- It has a response time of approximately 2–3 data updates per second without interrupting the analog signal.
- Masters are connected parallel to field devices, So devices can be easily connected or disconnected and the current loop remains uninterrupted.

Frequency Shift Keying



HART Networks

- There are two types of HART networks:

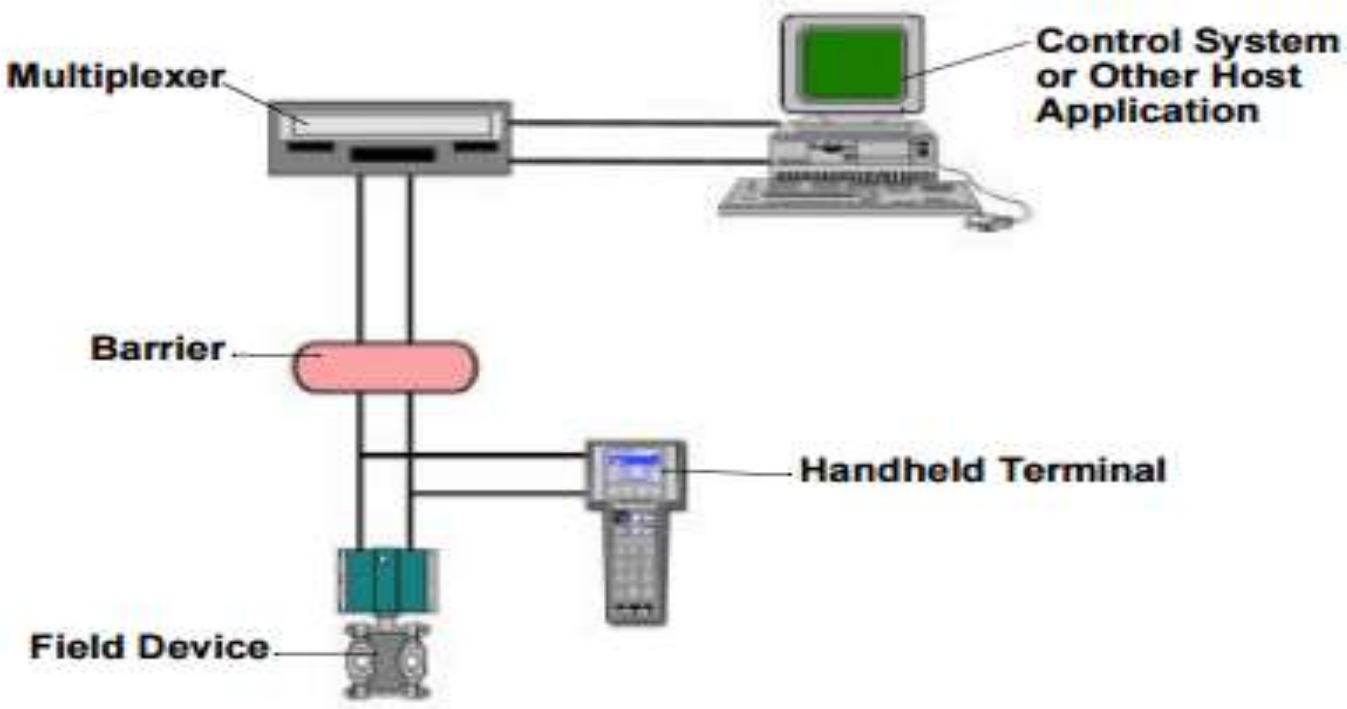
1. **Point-to-Point Mode**

   Here, the traditional 4–20 mA signal is used to communicate one process variable, while additional process variables, configuration parameters, and other device data are transferred digitally using the HART protocol.

   The HART communication digital signal gives access to secondary variables and other data that can be used for operations, commissioning, maintenance, and diagnostic purposes.

   The 4–20 mA analog signal is not affected by the HART signal and can be used for control in the normal way.

Point to Point Mode of Operation



Note: Instrument power is provided by an interface or external power source that is not shown.

HART Networks

2. **Multidrop mode**

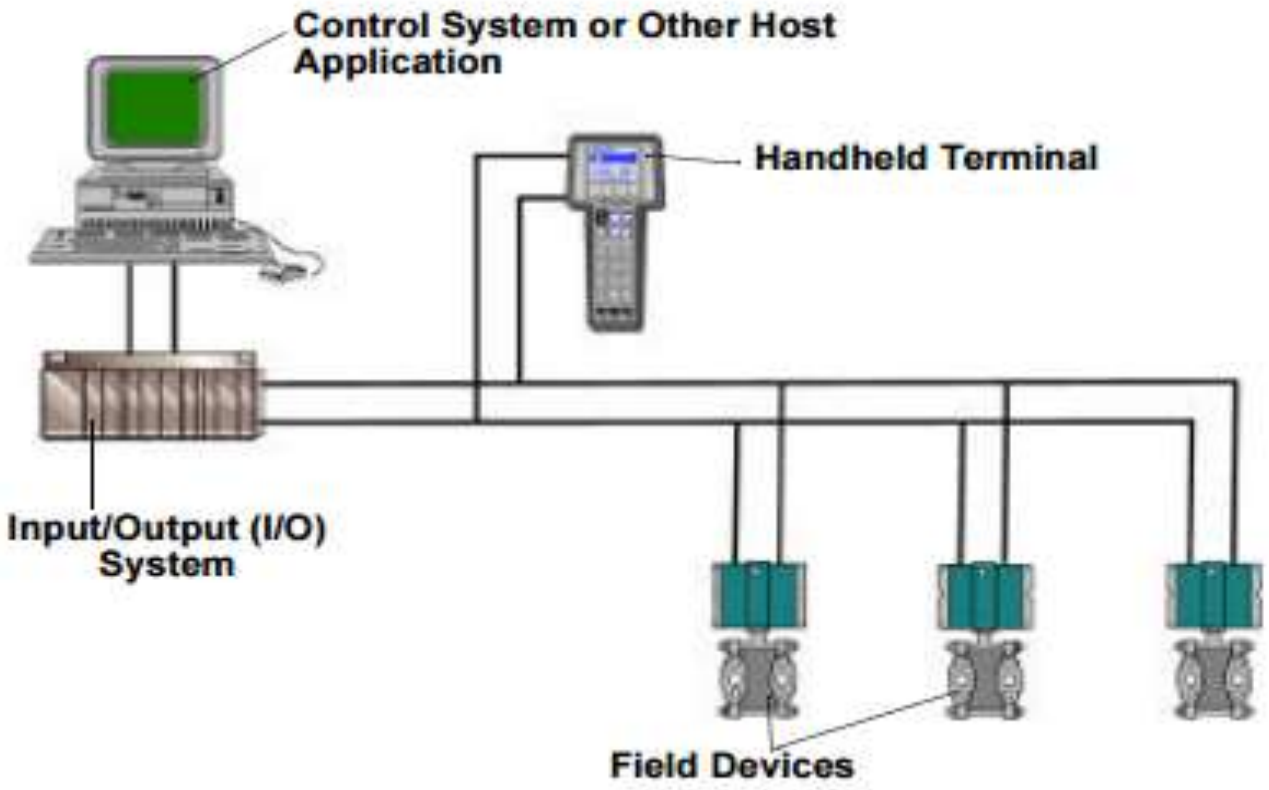
In multidrop operation, the devices exchange their data and measured values only via the HART protocol. The analog current signal serves just to energize the two-wire devices, providing a direct current of 4 mA.

Upto 15 field devices are connected in parallel to a single wire pair and the host distinguishes the field devices by their preset addresses which range from 1 to 15.

The multidrop mode of operation requires only a single pair of wires and, if applicable, safety barriers and an auxiliary power supply for up to 15 field devices . All process values are transmitted digitally. In multidrop mode, the current through each device is fixed to a minimum value (typically 4 mA).

Generally, multidrop connection is used for supervisory control installations that are widely spaced, such as pipelines, custody transfer stations, and tank farms.

Multidrop Mode of Operation



Note: Instrument power is provided by an interface or external power source that is not shown.

HART commands

- The HART command set provides uniform and consistent communication for all field devices.
- Command set includes three classes:

1. Universal

   All devices using the HART protocol must recognize and support the universal commands. Universal commands provide access to information useful in normal operations (e.g., read primary variable and units).

2. Common Practice

   Common practice commands provide functions implemented by many, but not necessarily all, HART communication devices.

HART commands

3. Device Specific

Device-specific commands represent functions that are unique to each field device. These commands access setup and calibration information, as well as information about the construction of the device. Information on device-specific commands is available from device manufacturers.

HART Commands

Universal Commands	Common Practice Commands	Device-Specific Commands
• Read manufacturer and device type • Read primary variable (PV) and units • Read current output and percent of range • Read up to four predefined dynamic variables • Read or write eight-character tag, 16-character descriptor, date • Read or write 32-character message • Read device range values, units, and damping time constant • Read or write final assembly number • Write polling address	• Read selection of up to four dynamic variables • Write damping time constant • Write device range values • Calibrate (set zero, set span) • Set fixed output current • Perform self-test • Perform master reset • Trim PV zero • Write PV unit • Trim DAC zero and gain • Write transfer function (square root/linear) • Write sensor serial number • Read or write dynamic variable assignments	• Read or write low-flow cut-off • Start, stop, or clear totalizer • Read or write density calibration factor • Choose PV (mass, flow, or density) • Read or write materials or construction information • Trim sensor calibration • PID enable • Write PID setpoint • Valve characterization • Valve setpoint • Travel limits • User units • Local display information

WIRING AND INSTALLATIONS

- WIRING

  HART wiring in the field usually consists of twisted pair cables.

  If very thin and/or long cables are used, the cable resistance increases so signal attenuation and distortion increases.

  For trouble-free transmission, the cables must have a sufficient cross section and an appropriate length.

  If interference signals are a problem, long lines must be shielded.

  The signal loop and the cable shield should be grounded at one common point only.

WIRING

- According to the specification, the following configurations work reliably:
- For short distances, simple unshielded 0.2 mm^2 two-wire lines are sufficient.
- For distances of up to 1,500 m, individually twisted 0.2 mm^2 wire pairs with a common shield over the cable should be used.
- For distances of up to 3,000 m, individually twisted 0.5 mm2 two-wire lines shielded in pairs are required.

WIRING AND INSTALLATIONS

- PLUG CONNECTORS

  An essential benefit is that HART integrates the existing wires. So the HART specification does not prescribe the use of a specific type of plug connector.

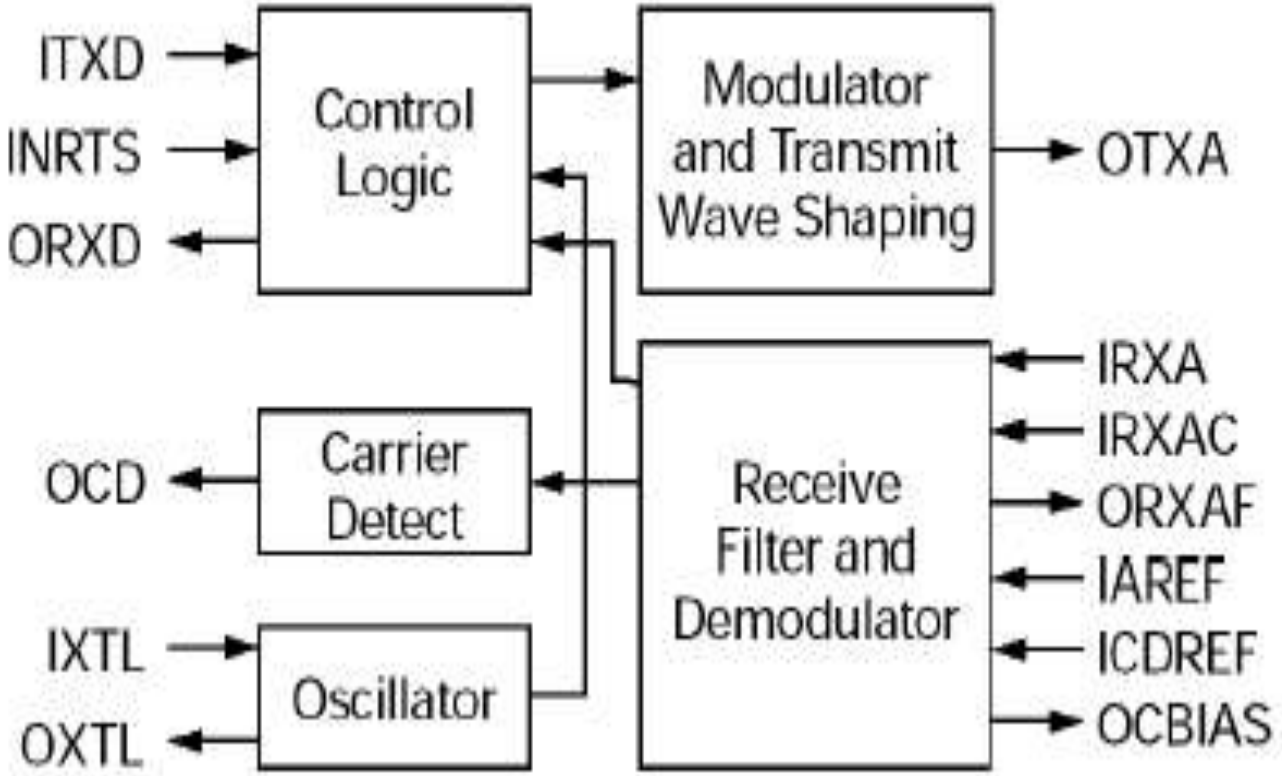
  HART signals are usually connected via simple clamp terminals.

- HART-COMPATIBLE FEATURES

  HART communication between two or more devices can function properly only when all communication participants are able to interpret the HART sine wave signals correctly.

  Inputs and outputs with an internal resistance that falls below the FSK frequency range short-circuit the HART signals. To prevent this, the internal resistance must be increased using an additional circuit, RC low pass (250 Ω, 1 µF) circuit.

OPERATION



A general 20C15HART Modem

Pinout of 20C15 HART Modem

Pin Number	Name	In/Out	Description	Comment
1,5,7,8,9,14			Test	Connect To Ground
2,3,4,25,28			Test	Leave Open
6	/RESET	In	Reset (Low To Reset)	Connect To R & C
10	OTXA	Out	Transmit Out	Analog FSK Hart signal Out
11	IAREF	In	Reference Voltage	Connect to 1.235V for 3.3V supply, Connect to 2.5V for 5.0V supply.
12	ICDREF	In	Carrier Detect Reference Voltage	Set at 80 mV below IAREF
13	OCBIAS		Comparator Bias Current	Connect R_{BIAS} to GND to obtain $\frac{V_{IAREF}}{R_{BIAS}} = 2{,}5\mu A$
15,22			Power Pin	Connect to 3.3 volt or 5.0 volt.
16	IRXA	In	Input to Receive Gain Stage	Analog FSK Hart signal In
17	ORXAF	Out	Output from Rcv Gain Stage	Low Impedance Point
18	IRXAC	In	Receive Filter Continued	
19	OXTL	Out	Crystal Oscillator Output	460.8 kHz or External Clock
20	IXTL	In	Crystal Oscillator Input	XTAL or Control External Clock
21			Ground	
23	INRTS	In	Request-to-Send	Selects between XMIT and RCV
24	ITXD	In	Transmit Data	
26	ORXD	Out	Receive Data	
27	OCD	Out	Carrier Detect	

Modulator

- **Modulator (Transmitter) :**
- The modulator is operating (and the demodulator is shut down) whenever INRTS (request to send) is low.
- When ITXD (transmit data) is high the modulator output at OTXA(transmit out) will be a trapezoidally shaped wave at nominally 1200 Hz.
- When ITXD is low the modulator output is nominally 2200 Hz.
- OTXA is usually AC-coupled to an amplifier or buffer stage.
- The output voltage levels at OTXA depend on the reference voltage applied at IAREF.

Demodulator

Demodulator (Receiver) :

- The demodulator is operating (and the modulator is shut down) whenever INRTS is high.
- The received signal is first applied to a band pass filter. Part of this filter is off-chip to reduce interference.
- Other pins that are part of the receive filter are IRXA, ORXAF, and IRXAC.
- The filter output is applied to two comparators --

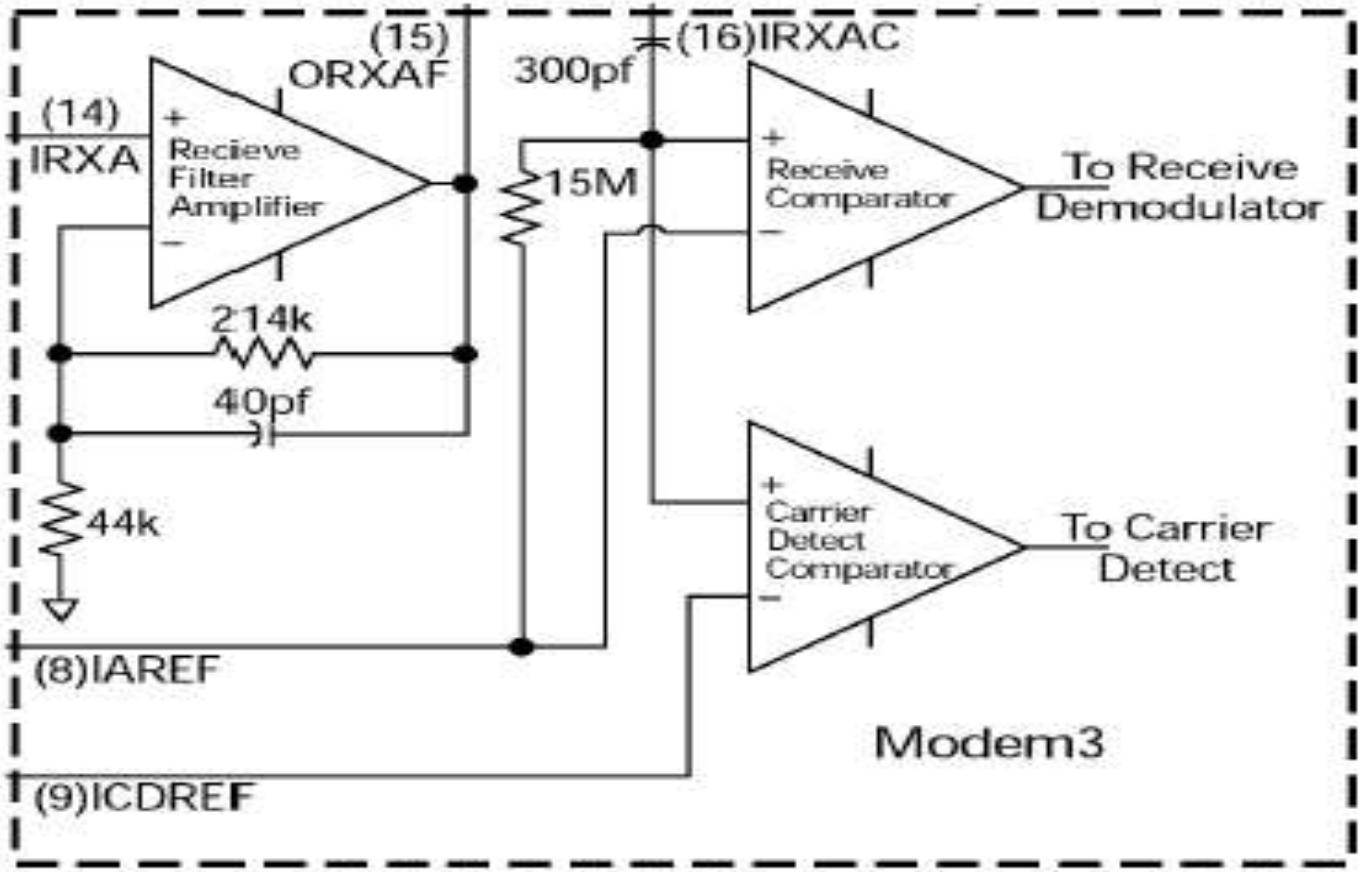
  one to slice the signal and produce a logic-level version of the received FSK

  and second to act as a carrier detect.

Demodulator

- The reference for the first comparator is the reference voltage applied at IAREF and for the second is applied at ICDREF and is normally set to be 80mV below the level at IAREF.
- Logic circuits following the carrier detect comparator are used to decide whether carrier is present or not.
- The logic-level FSK out of the first comparator is applied to a logic block that generates a high level if a frequency below 1700 Hz is present and a low level if a frequency above 1700 Hz is present.
- The output is given to UART for decoding.

Demodulator



Operation

- **Parameters:**
- **Clock:** 460.8 kHz, Tolerance 1%.
- **Power Supply Current:** 400 µA max. (3.3 volt supply), 600 µA max. (5.0 volt supply).
- (This is the current during receive. It is about 100 µA to 200 µA less during transmit)
- **Operating Temperature Range**: 0°C to 70°C.
- **Reset Minimum Pulse Time**: 2 µS
- **Transmit Output Drive Capability**: Needs minimum of 30 k.
- **Circuits** : Field Instrument, 5.0 Volt Supply:

Benefits of HART Communications

- HART protocol provides a unique communication solution that is backward compatible with the installed base of instrumentation in use today. This backward compatibility ensures that investments in existing cabling and current control strategies will remain secure well into the future.
- Benefits outlined in this section include:

✓ Improved plant operations

✓ Operational flexibility

✓ Instrumentation investment protection

✓ Digital communication

Improved Plant Operations

- The HART protocol improves plant performance and provide savings in:

  - Commissioning and installation

  HART-based field devices can be installed and commissioned in a fraction of the time required for a traditional analog-only system.

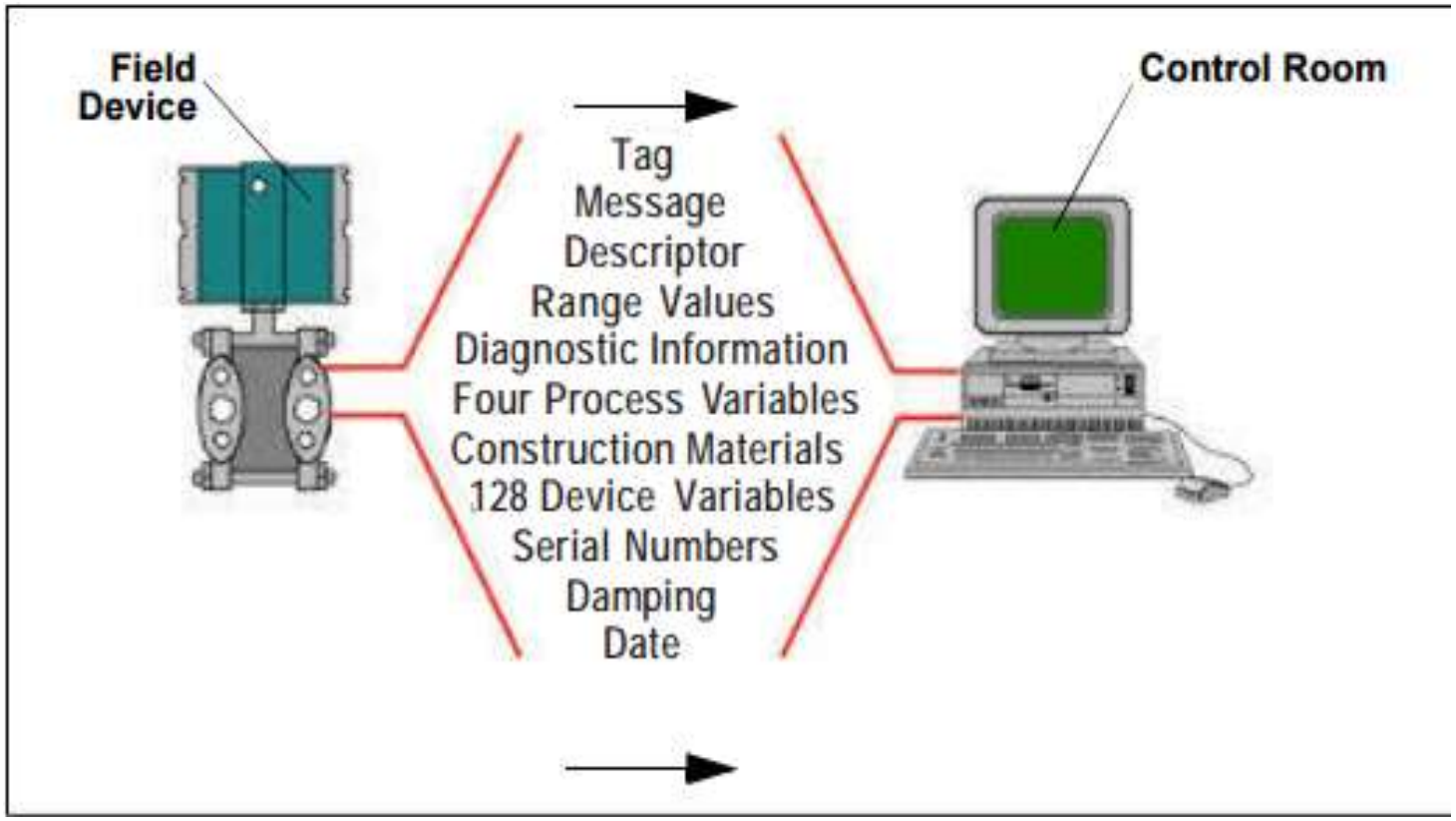
  - Plant operations and improved quality

. Provides accurate information that helps improve the efficiency of plant operations. During normal operation, device operational values can be easily monitored or modified remotely. If uploaded to a software application, these data can be used to automate record keeping for regulatory compliance.

  - Maintenance

  The HART protocol supports the networking of several devices on a single twisted wire pair. This configuration can provide significant savings in wiring and maintenance, especially for applications such as tank monitoring.

Improved Plant Operations

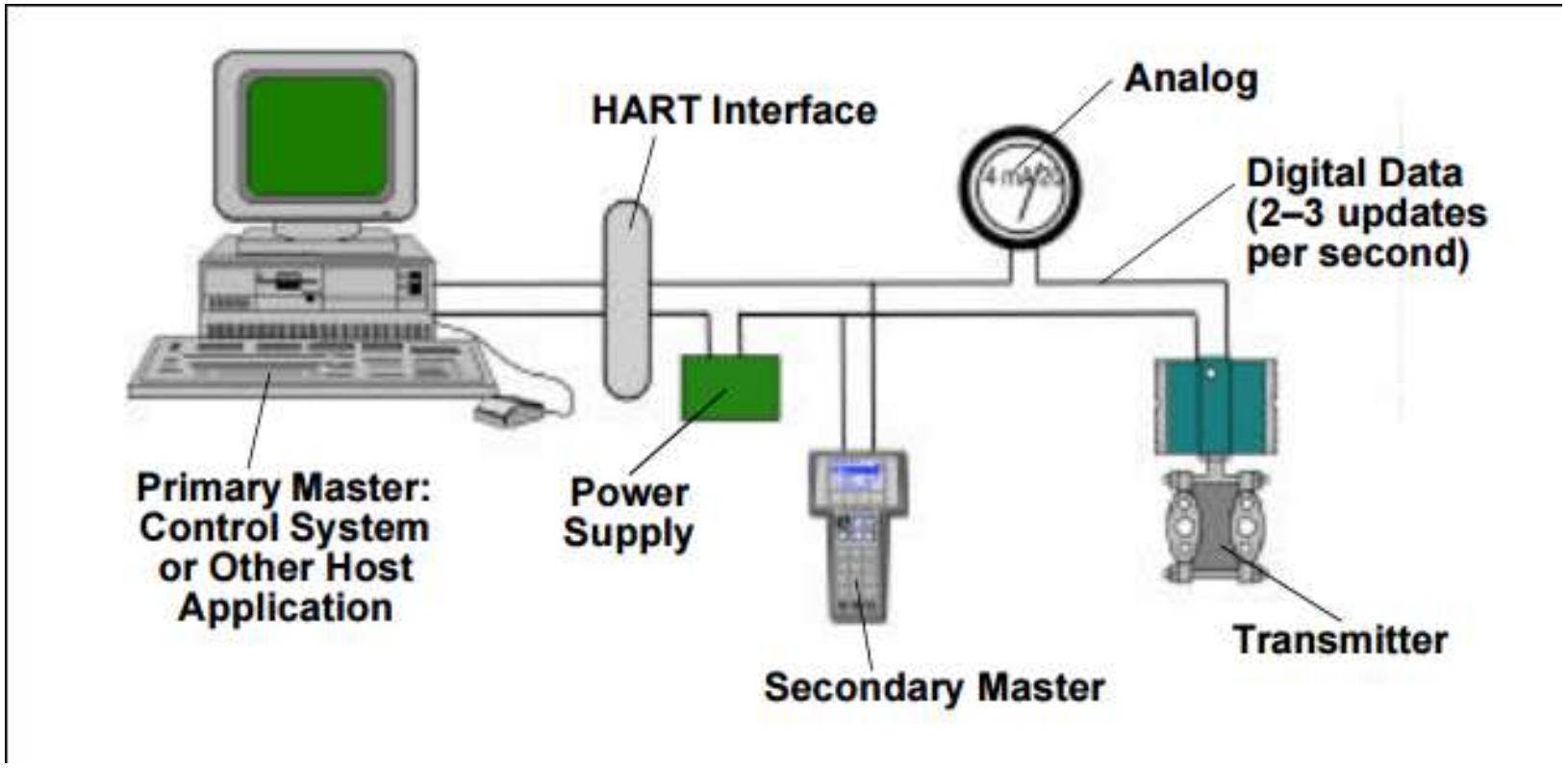


Examples of Device Parameters sent to control room

Operational Flexibility

- The HART protocol allows two masters (primary and secondary) to communicate with slave devices and provide additional operational flexibility. A permanently connected host system can be used simultaneously, while a handheld terminal or PC controller is communicating with a field device.
- The HART protocol ensures interoperability among devices through universal commands that enable hosts to easily access and communicate the most common parameters used in field devices.
- DDL enables a single handheld configuration or PC host application to configure and maintain HART-communicating devices from any manufacturer.
- It minimizes the amount of equipment and training needed to maintain a plant.

Multimaster System



Instrumentation Investment Protection

- Existing plants and processes have considerable investments in wiring, analog controllers, junction boxes, or smart instrumentation. The people and equipment already exist for the support and maintenance of the installed equipment. HART field instruments protect this investment by providing compatible products with enhanced digital capabilities
- Advanced installations can also use control systems with HART I/O capability. The status information can be used directly by control schemes to trigger remedial actions and allow on-line rearranging based on operating conditions and direct reading of multivariable instrument data.
- As HART field devices are upgraded, new functions may be added. A basic premise of the HART Protocol is that new HART instruments must behave in precisely the same manner as older versions when interfaced with an earlier revision host system.

Digital Communication

- A digital instrument that uses a microprocessor provides many benefits. These benefits are found in all smart devices regardless of the type of communication used. A digital device provides advantages such as improved accuracy and stability.
- The HART protocol enhances the capabilities of digital instruments by providing communication access and networking.

Digital Communication

Benefits	HART Instruments	Digital Instruments
Accuracy and stability	✓	✓
Reliability	✓	✓
Multivariable	✓	✓
Computations	✓	✓
Diagnostics	✓	✓
Multiple sensor inputs	✓	✓
Ease of commissioning	✓	
Tag ID	✓	
Remote configuration	✓	
Loop checks	✓	
Adjustable operational parameters	✓	
Access to historical data	✓	
Multidrop networking	✓	
Access by multiple host devices	✓	
Extended communication distances	✓	
Field-based control	✓	
Interoperability	✓	

CONCLUSION

- HART protocol provides a unique communication solution that is backward compatible with the installed base of instrumentation in use today. This backward compatibility ensures that investments in existing cabling and current control strategies will remain secure well into the future.
- The HART communication protocol is based on the Bell 202 telephone communication standard and operates using the frequency shift keying (FSK) principle. The digital signal is made up of two frequencies— 1,200 Hz and 2,200 Hz representing bits 1 and 0, respectively.
- Versatile-It continuously validates the integrity of control information.
- It ensures accuracy of system data and detect any deviation between device and system.
- It gives additional information in multi variable devices.

References

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Thank you!!

Questions??